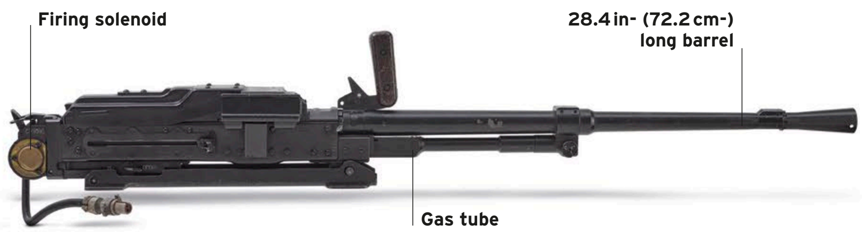
Firepower

In tank warfare, the shape of things to come was sketched out near Villers-Bretonneux on April 24, 1918, when British and German tanks met for the first time. The British prevailed, thanks to one of their vehicles being a "male," armed with two 6-pounder QF guns. During the interwar period, however, tank-on-tank encounters were not uppermost in the minds of designers or strategists, and it took exposure to a new type of mechanized warfare during World War II to shake the belief that the primary role of the tank was to support infantry. This remained important, but as tank armour grew thicker, guns and ammunition had to grow increasingly specialized in order to reliably penetrate it. In 1945 most tank guns firing AP rounds had muzzle velocities of around 2,800ft/s (850m/s), and could penetrate roughly 6–8in (150–200mm) of armour at 328ft (100m). By 2010 this had increased to over 5,750ft/s (1,750m/s) with APFSDS, giving penetration of over 23.6in (600mm) at 6,560ft (2,000m).

MACHINE GUNS

Tanks will always be vulnerable at close quarters against determined infantry, with machine guns being the usual defence. Most modern tanks mount at least two—one coaxially (i.e on the same axis) with the main gun, and one mounted on the roof that is aimed independently. Up until the late 1940s, most tanks also had a bow machine gun in the front of the hull. This provided extra firepower, but was difficult to aim. It also created a weak point in the frontal armor. As main-gun ammunition increased in size, the space was instead used to store more of it. Coaxial and bow machine guns are usually of around 7.62 mm/0.3 in caliber. Roof-mounted guns often fire heavier 12.7 mm/0.5 in rounds. On some tanks this weapon can be aimed and fired from inside the vehicle.

Vickers Mark VI .303 in machine gun
Variants of the Vickers machine gun, including the Mark VI, were used as a secondary armament in a number of British tanks during the interwar period. They were gradually replaced in tanks by Browning and Besa machine guns in the early 1940s, although the Vickers continued to be used elsewhere until the 1960s.



PKT 7.62mm machine gun
The PKT was developed by Mikhail Kalashnikov from his AK assault rifle, but chambered for the longer and more powerful 7.62 x 54 mm rimmed round. As it was mounted coaxially, the sights, butt, bipod, and trigger were not fitted. Instead, an electrically fired solenoid trigger unit was installed and the tanks' sights were used for aiming.

Browning M2 .50-caliber machine gun
One of a number of highly reliable recoil-operated designs developed by John Moses Browning, the M2 has been used by infantry, on armored and unarmored vehicles, aboard ship, and on aircraft since the 1920s. When fitted to a tank, it is invariably roof-mounted and aimed by the commander.

MAIN GUNS

The development of the tank's main gun has been largely linear. Size, both in terms of caliber and barrel length, has steadily increased in order to fire more powerful ammunition, but the fundamental principle of a high-velocity, direct-fire weapon remains. Many of the innovations in tank gunnery have been in fire control systems, ensuring that this weapon can hit its target as often as possible. Modern systems integrate stabilizers, laser range finders, high-magnification thermal sights, and ballistic computers to allow highly accurate fire at extreme range under any conditions. Another innovation is the autoloader, which uses a mechanical system rather than a crew member to select and load ammunition. Many recent tanks are armed with smoothbore guns, which fire projectiles stabilized by fins rather than spinning. Smoothbores can also be used to fire guided missiles.